BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) (33)

Maximum Marks: 33

Note: Select the correct answer for each from the given options. (Options in the original are separated by star marks, shown here as A–D in printed order. Numbering xxxvii/xxxviii follows the original print.)

Note: Time: 3 Hours — Maximum Marks: 85

Q1. Select the correct answer for each from the given options:

Qi. The value of absolute zero on the Fahrenheit scale is:

(A) 491.[illegible in source scan]°F
(B) −459.4°F
(C) −523.4°F
(D) 459.4°F

Qii. The resistivity of a material is ρ .If area of cross-section of material is doubled and length is halved then the resistivity of conductor:

(A) remains constant
(B) increases 4 times
(C) increases 2 times
(D) is one fourth

Qiii. A voltmeter is an ideal one if its internal resistance is:

(A) very large
(B) large
(C) small
(D) infinite

Qiv. Pentavalent impurities are also known as:

(A) Accepter Impurities
(B) Donor Impurities
(C) Hole
(D) Electron

Qv. The relativistic mass of 5kg ball observed by observer moving with a speed equal to speed of light is:

(A) 5kg
(B) 10 kg
(C) infinite
(D) 20 kg

Qvi. An intense, monochromatic, narrow and coherent beam of light is called:

(A) X − rays
(B) β − rays
(C) LASER
(D) MASER

Qvii. The force that holds nucleons together in nucleus is:

(A) nuclear force
(B) coulombs force
(C) gravitational force
(D) weak force

Qviii. This is the most penetrating:

(A) α - particle
(B) β - particle
(C) γ − rays
(D) positron

Qix. Four bulbs of 10W, 20W, 30W and 40W are connected in series. The bulb that will shine the most is:

(A) 10W
(B) 20 W
(C) 30 W
(D) 40W

Qx. Photoelectric effect proves this nature of light:

(A) Wave
(B) Particle
(C) Polarization
(D) Diffraction

Qxi. If the number of electrons in excited state exceeds than the number of electrons in the ground state, then this will occur:

(A) Excitation Potential
(B) Population Inversion
(C) Excitation Energy
(D) Ionization Energy

Qxii. A current of 9 amp flows in a circuit of three resistors connected parallel. If each resistor has magnitude 5 Ω, then current in the 3rd resistor is:

(A) 9 amp
(B) 3 amp
(C) 1.5 amp
(D) 0.5 amp

Qxiii. If the forbidden energy gap is very small, then material is:

(A) Insulator
(B) Conductor
(C) Semi-Insulator
(D) Semi-Conductor

Qxiv. In this process, matter is converted into energy:

(A) Pair production
(B) Compton effect
(C) Photo electric effect
(D) Annihilation of matter

Qxv. LASER produces three dimensional image called:

(A) monogram
(B) spectrogram
(C) hologram
(D) photogram

Qxvi. The number of neutrons in $_{92}U^{235}$ is:

(A) 92
(B) 146
(C) 143
(D) 140

Qxvii. A transformer step down:

(A) Energy
(B) AC only
(C) DC only
(D) both AC & DC

Qxviii. The section of transistor that supplies charge carriers (electrons or holes) is called:

(A) collector
(B) base
(C) emitter
(D) junction

Qxix. In pair production the electron and positron move in opposite direction to conserve:

(A) Momentum
(B) Charge
(C) Energy
(D) Mass

Qxx. If distance between the plates of capacitor of capacitance C is doubled and area is halved, the capacitance will become:

(A) C/4
(B) C
(C) 4C
(D) 2C

Qxxi. Induced emf cannot be produced by:

(A) change of flux
(B) movement of a coil
(C) AC source
(D) DC source

Qxxii. A thermodynamic process, such as compression or expansion done very quickly, will nearly be:

(A) Isothermal
(B) Adiabatic
(C) Isochoric
(D) Isobaric

Qxxiii. An electron of mass 'm' and charge 'e' is accelerated by a difference of potential V, its speed will be:

(A) $v = \sqrt{(2Ve/m)}$
(B) $v = \sqrt{(2Vm/e)}$
(C) $v = \sqrt{(2Vme)}$
(D) $v = \sqrt{(m/2Ve)}$

Qxxiv. Electric field intensity at a point inside a uniformly charged hollow sphere is given by:

(A) $\sigma/2\varepsilon_0$
(B) σ/ε_0
(C) $2\varepsilon_0/\sigma$
(D) zero

Qxxv. If work done in moving positive point charge along a surface is zero then this surface must be:

(A) Hypothetical
(B) Irregular
(C) Equipotential
(D) Rough

Qxxvi. A particle of mass and charge is to be held motionless by an electric field E, the electric intensity will be:

(A) E = mg/g
(B) E = mg/q
(C) E = mgq
(D) E = q/mg

Qxxvii. A 12 volt, battery with internal resistance 0.1Ω is to be charged at 10 Amp, the charger must be of this volt:

(A) 11
(B) 9
(C) 13
(D) 12

Qxxviii. A 2.2kW electric iron operates at 220 volt, the current it draws:

(A) 20 Ampere
(B) 22 Ampere
(C) 10 Ampere
(D) 5 Ampere

Qxxix. Lenz's law is direct consequence of this:

(A) Ohm's law
(B) Coulomb's law
(C) Faraday's law
(D) law of conservation of energy

Qxxx. When a charged particle enters in a magnetic field $\vec{B}$ with velocity $\vec{v}$ making angle θ, the trajectory of particle will be:

(A) circular
(B) elliptical
(C) helix
(D) straight line

Qxxxi. The minimum energy required for pair production is:

(A) 2.01 MeV
(B) 1.02 MeV
(C) 0.21 MeV
(D) 1.20 MeV

Qxxxvii. In 'L' shell of hydrogen atom, angular momentum of an electron on is:

(A) h/2π
(B) 2h/π
(C) h/π
(D) 2πh

Qxxxviii. Isotope ${}_{6}C^{14}$ is mostly used in:

(A) Industry
(B) Medical Science
(C) Archeology
(D) Agriculture

Section 'B' (Short-Answer Questions) (32)

Maximum Marks: 32

> Note: Answer any EIGHT part questions from this section. All questions carry equal marks. Draw diagrams where necessary.

Q2.

(i) A Carnot engine works between 800°C and 400°C, if source temperature is increased by 50°C or sink temperature is decreased by 50°C, then which will cause greater efficiency.

(ii) What amount of current is required to produce a magnetic field of $3\sqrt{2}$ Tesla at a distance of 2 cm from a long straight current carrying wire? ($\mu_0 = 4\pi \times 10^{-7}$ Weber/Am)

(iii) In photo electric experiment, a stopping potential of 1.25 volt is needed. Find the maximum kinetic energy and maximum speed of photo electron e = 1.6 x 10^{-19}C ; m_e = 9.1 x 10^{-31} kg

(iv) Write a note on Wilson Cloud Chamber.

(v) What is semi conductor diode? Describe the working of full wave rectifier using semiconductor diode with the help of a circuit diagram.

(vi) Define electric potential difference (V) and electric field intensity ($\vec{E}$). Derive relation $V = \vec{E}.\vec{d}$)

(vii) A charged particle of charge 2 x10^{-6}C, in an electric field between two parallel metal plates 4 cm apart, is acted upon by a force of 10^{-4}N a) What is the intensity of the electric field? b) What is potential difference between the plates,

(viii) On what factors resistance of conductor at given temperature depends, Derive relevant formula.

(ix) A 400 volt voltmeter has a total resistance of 40,000 Ω. What additional series resistance mug be connected to it to increase the range to 750 volts?

(x) Hydrogen atom in ground state is excited by absorbing a photon of 12.15eV. Find the quantum number of this excited state.

(xi) If the number of atoms per gram of $_{88}Ra^{226}$ is 2.666 x 10^{21} and it decays with the half life of 1622 year. Find decay constant and activity of the sample.

(xii) Who will be the velocity and momentum of panicle whose rest mass is m_0 and whose K.E. (Kinetic Energy) is equal to its rest mass energy.

Section "C" (Detailed Answer Questions) (20)

Maximum Marks: 20

> Note: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

Q3.

(a) State Ampere's law. Apply this taw to find the magnetic field of induction inside a solenoid.

(b) Explain Compton Effect and obtain an expression for the increase in the wavelength of the scatter photon.

Q4.

(a) Describe the construction and working of a moving coil Galvanometer. Show that the deflection produced in galvanometer is directly proportional to the current.

(b) Derive an expression for total energy of the electron in the nth orbit of a hydrogen atom.

Q5.

(a) Derive pressure formula of an ideal gas using kinetic molecular theory.

(b) State Gauss's law and write mathematical expression. Apply this law to determine the electric field at a point close to a thin, infinite sheet of positive charges.